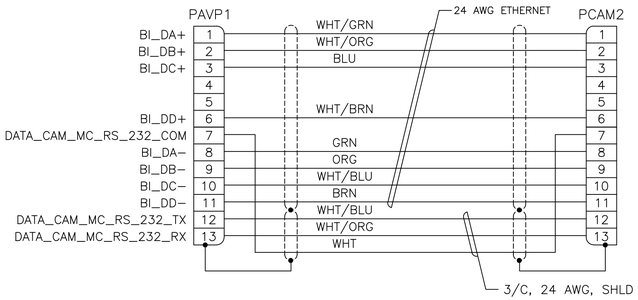
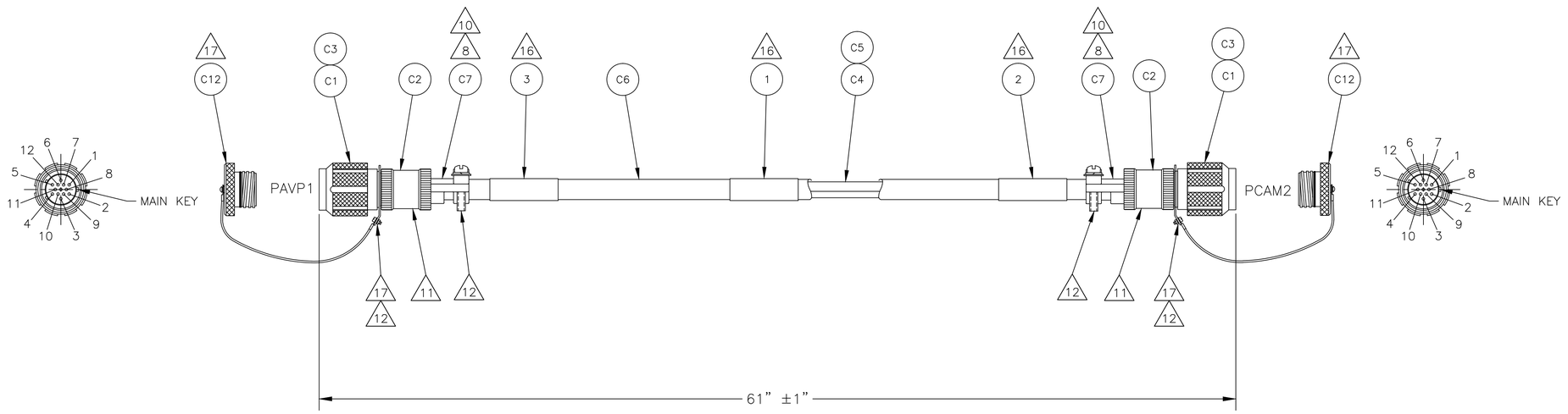
COMMERCIAL PARTS LIST

ITEM	QTY	PART NUMBER	DESCRIPTION	MANUFACTURER
C1	2	805-002-16UC10-13SA	CONNECTOR, CIRCULAR, 13 POS.	GLENAIR
C2	2	380MS137UC0806-6	BACKSHELL, STRAIGHT	GLENAIR
C3	A/R	809-002	CONTACT, SOCKET, SIZE 23, AWG 22-28	GLENAIR
C4	A/R	E50824	WIRE, 24 AWG, ETHERNET, 4 PAIR	PIC WIRE & CABLE
C5	A/R	M27500-24RC3S06	3/C, 24 AWG TWISTED, SHIELDED JACKETED	QPL
C6	A/R	5011001332S	EXPANDABLE BRAIDED SLEEVING, 1/2"	BENTLEY HARRIS
C7	A/R	THFE4-500015	SHRINK TUBING, EXPANDED PREHEAT ID .50"	SAINT GOBAIN
C8	A/R	P-224X.5	TAPE, KAPTON, 1/2" WIDE	PERMACEL
C9	A/R	222MS	THREADLOCKER	LOCTITE
C10	A/R	7649	PRIMER N	LOCTITE
C11	A/R	2216 B/A GRAY	EPOXY, 2 PART	3M
C12	2	667-261-UC-H1054-5	CONN. CAP, SIZE 10, 5" TETHER/SPLIT RING	GLENAIR

DRAWING LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	S1101EFT2002-0X	COMBO LABEL
2	1	S1101EFT2002-03	CONNECTOR ID
3	1	S1101EFT2002-04	CONNECTOR ID



SCHEMATIC



14. TEST PER NASA-STD-8739.4 PARA 18.2.6 USING THE FOLLOWING:
    CONTINUITY: < 5 OHMS
    DWV: VOLTAGE: 750VAC ±50VAC
    CURRENT: < 1 MILLIAMP
    DURATION: < 1 MINUTE
    IR: VOLTAGE: 500VDC ±50VDC
    RESISTANCE: > 100 MEG OHMS
    DURATION: < 1 MINUTE
15. PLACE COMPLETED ASSEMBLY IN CLEAN, STATIC-SHIELDING PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER/STATEMENT OF WORK PER SAE AS478, METHOD 35D. STORE IN A PROJECT APPROVED STORAGE AREA.

△16 APPLY LABELS, ITEMS 1-3 TO CLEAN, DRY SURFACE IN LOCATION SHOWN.

△17 APPLY EPOXY (ITEM C11) TO WIRE TETHER ENDS.

NOTES:

1. MATERIAL CERTIFICATION REQUIRED PER QUALITY LEVEL SPECIFIED IN WORK ORDER/SOW.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. HARNESS ASSEMBLY AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.4 AND FOLLOWING NOTES:
   a. TO PREVENT POTENTIAL DAMAGE (COLD FLOW) OF THE UNDERLYING WIRE INSULATION, SEPARATOR SHALL BE APPLIED OVER THE WIRE BUNDLE TO GIVE THE WIRE CONTINUOUS PROTECTION.
   b. ALL SHIELD TERMINATIONS ARE INTERFERENCE SENSITIVE.
5. FOR TENSILE PULLS SEE TENSILE PULL STRENGTH LIST. TENSILE PULL STRENGTHS PER NASA-STD-8739.4 AND/OR THE MANUFACTURER'S SPECIFICATIONS.
6. INSTALL CONTACTS IN ALL UNUSED POSITIONS OF CIRCULAR CONNECTORS.
7. PERFORM CONTACT RETENTION TEST PER NASA-STD-8739.4 AND/OR MANUFACTURER'S SPECIFICATIONS. CONTACTS REQUIRING A PULL TEST FOR RETENTION SHALL BE PERFORMED AS FOLLOWS: A GENTLE PULL ON THE WIRE SHALL BE USED TO DETERMINE IF THE CONTACT IS SEATED PROPERLY.

△8 USE KAPTON TAPE (ITEM C8) TO SECURE EXPANDO SLEEVE (ITEM C6) WITHIN BACKSHELL.

9. TERMINATE SHIELDS INTO BACKSHELL PER MANUFACTURER'S SPECIFICATION.

△10 TO SAFELY SECURE CABLE IN STRAIN RELIEF INCREASE THE CABLE DIAMETER USING HEAT SHRINK TUBING (ITEM C7). MAXIMUM OF TWO (2) LAYERS.

△11 TORQUE BACKSHELL (ITEM C2) TO 20 ±2 IN-LBS. STAKE BACKSHELL USING EPOXY (ITEM C11).

△12 TRIM FASTENERS TO EXPOSE 1-2 THREADS. APPLY PRIMER N (ITEM C10) AND THREADLOCKER 222MS (ITEM C9), TORQUE FASTENERS PER CHART. APPLY EPOXY (ITEM C11) TO EXPOSED THREADS.

13. -REMOVED-

CRIMP TOOL LIST			
CONTACT P/N	CRIMP TOOL	POSITIONER	I/E TOOL
809-002	M22520/2-01	K1461	809-088

TENSILE PULL STRENGTH LIST			
MANUFACTURER	CONTACT P/N	WIRE AWG	LBS
GLENAIR	809-002	24	8

BACKSHELL CLAMPING TORQUE TABLE		
SCREW SIZE	TORQUE RANGE	UNITS
2-56	2 ±0.5	IN-LB
4-40	4 ±0.5	IN-LB
6-32	6 ±1.0	IN-LB
8-32	8 ±1.0	IN-LB
10-32	10 ±1.0	IN-LB
.250-20	12 ±1.0	IN-LB

REV	DESCRIPTION	APPROVED
C	REVISED PER ACME-ECO-527 AND ACME-ECO-533.	MH 4/11/16 DS 4/11/16 TG 4/21/16
B	REVISED PER ACME-ECO-486 AND ACME-ECO-501.	MH 8/26/15 DS 8/27/15 TG 8/31/15
A	REVISED PER ACME-ECO-276.	MH 12/15/14 TG 12/15/14 RP 12/23/14

REVISION HISTORY

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. ALL DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCE ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.
APPROVED: E. McCLOSKEY 4/21/16
PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	M. HOMKO 9/3/13	QUALITY	D. SEELEY 9/4/13
REVIEWER	J. BRUEWER 9/5/13	MATERIALS	M. WILKINSON 9/4/13
MECHANICAL	A. DRAKE 9/4/13	PROJ LEAD	B. BOROWSKI 9/10/13
ELECTRICAL	T. GOBELI 9/10/13		
STRUCTURAL	C. WERNER 9/4/13		
SAFETY	W. BORRELLI 9/3/13		

DIMENSIONS AND TOLERANCING PER ASME Y14.5M-1994

ZIN TECHNOLOGIES
6745 ENGLE ROAD
AIRPORT EXECUTIVE PARK
CLEVELAND, OHIO 44130
(440) 625-2200
www.zin-tech.com

CONTRACT NUMBER: NNC09BA02B
PROJECT CODE: SPACE110

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
JOHN H. GLENN RESEARCH CENTER
LEWIS FIELD
CLEVELAND, OHIO

TITLE
AVP TO DATA CAMERA HARNESS ASSEMBLY

SIZE	CAGE CODE	DWG NO	REV
D	0SG68	S1101EFA2002	C

SCALE NONE | INITIAL REL DATE 09/10/13 | SHEET 1 OF 1

RELEASE LEVEL: RELEASED
MODEL NAME: S1101EFA2002_REV_C.DWG

AutoCAD 2012

DWG NO S1101EFA2002 | SH 1 | REV C